

GATE SOLUTION IN MINING ENGINEERING

(Containing Solved Questions 2015 - 2007)

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Gate solution -2015

Q.1 Out of support categories given for an underground coal mine, identify the 'active support'.

- (A) wire mesh (B) shotcrete
(C) fully grouted roof bolt (D) hydraulic prop

Answer (D)

Q.2 Massive sandstone in immediate roof delays the local fall in goaf of a coal mine. Under this condition, crushing of the pillars at outbye side is called

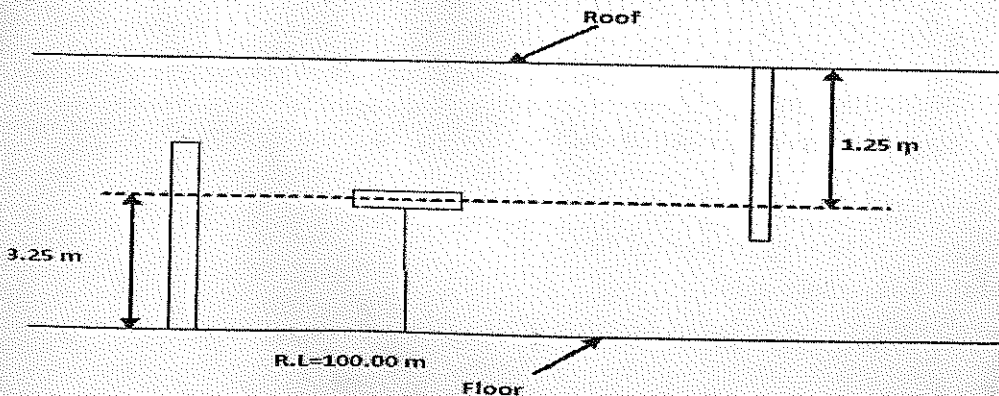
- (A) coal bump (B) overriding of pillars.
(C) stiffening of pillars (D) spalling of pillars

Answer (B)

Q.3 A back sight on a bench mark RL 100.00 m on the floor of a tunnel is 3.25 m. The inverse staff reading on a floor station of the tunnel is 1.25 m. The RL of the roof station in m is _____

Answer : 104 to 105 .

Solution .



$$\begin{aligned}\therefore \text{RL of the roof station is} &= 100 + 3.25 + 1.25 \\ &= 104.5 \text{ m}\end{aligned}$$

The angle in degrees at which a ridge line intersects contours is

- (A) 0 (B) 30 (C) 45 (D) 90

Answer (D)

In a drum hoisting system through a vertical shaft, overwinding is prevented by

- (A) Lilly controller (B) detaching hook
(C) caliper brake (D) safety catch

Answer (B)

The temperature of a parcel of air decreases from 30.2°C to 28.9°C as it rises from an altitude of 0 m to 120 m. The lapse rate for the the atmosphere is

- (A) subadiabatic (B) adiabatic (C) superadiabatic (D) transadiabatic

Answer (C)

on .

Lapse rate - The rate at which atmospheric temperature decreases with increase in altitude.

Superadiabatic lapse rate - This occurs when the atmosphere temperature drops more than $1^{\circ}\text{C}/100\text{ m}$.

The excess pore pressure in backfill material in a cut- and- fill stope leads to

- (A) reduction in the strength of the wall rock
(B) enhancement of bearing strength of fill
(C) loss of shear resistance of fill
(D) prevention of progressive failure of crown pillar

Answer (C)

The primary purpose of cut holes for blasting in an underground drivage is to

- (A) provide additional free face
- (B) have smooth surface after blasting
- (C) prevent over-breakage
- (D) reduce noise

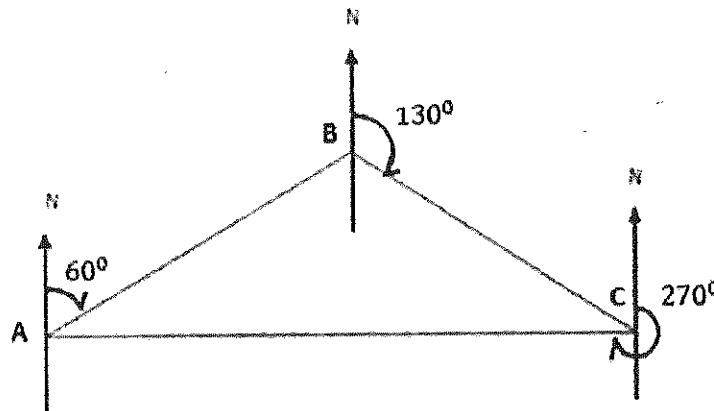
Answer (A)

Q.9 In a triangle ABC, the bearings of the sides AB, BC and CA are 60° , 130° , and 270° respectively. The interior angles A, B and C in degrees respectively are

- (A) 110, 40, 30
- (B) 40, 110, 30
- (C) 30, 40, 110
- (D) 30, 110, 40

Answer (D)

Solution .



$\therefore$ bearing of AB = 60°

$\therefore$ bearing of BA = $60^\circ + 180^\circ = 240^\circ$ and

$\therefore$ bearing of BC = 130°

$\therefore$ bearing of CB = $130^\circ + 180^\circ = 310^\circ$ and

$\therefore$ bearing of CA = 270°

$$\therefore \text{bearing of AC} = 270^\circ - 180^\circ = 90^\circ$$

Therefore ,

$$\angle A = \text{bearing of AC} - \text{bearing of AB}$$

$$\angle A = 90^\circ - 60^\circ = 30^\circ, \text{ and}$$

$$\angle B = \text{bearing of BA} - \text{bearing of BC}$$

$$\angle B = 240^\circ - 130^\circ = 110^\circ, \text{ and}$$

$$\angle C = 180^\circ - \angle B - \angle A = 180^\circ - 110^\circ - 30^\circ$$

$$\angle C = 40^\circ$$

Q.10 In a binomial distribution, the probability of success $p \rightarrow 0$ and number of trials $n \rightarrow \infty$ such that $\lambda = np$ approaches to a finite value. The variance of the distribution is

- (A) $np\lambda$ (B) $n\lambda$ (C) $p\lambda$ (D) λ

Answer (D)

Solution .

In binomial distribution : The variance of random variable X is given by -

$$\text{Var } [X] = np(1 - p)$$

$$\text{Since, } p \rightarrow 0, n \rightarrow \infty \text{ and } \lambda = np$$

Therefore,

$$\text{Var } [X] = \lambda(1 - 0)$$

$$\text{Var } [X] = \lambda$$

Q.11 For a function $f(x)$, it is given that $f(0) = 2$ and $f'(0) = 4$. Ignoring all other higher order derivative terms, the value of $f(0.5)$ is _____

Answer : 4.

Solution .

Suppose that, $f(x) = x^4 + 4x + 2$ (1)

When $x = 0$, then

$$f(x) = 2$$

Differentiating equation (1) w.r.t, x , then

$$f'(x) = 4x^3 + 4$$

When $x = 0$, then

$$f'(0) = 4$$

When $x = 0.5$, then from equation (1)

$$f(0.5) = 0.5^4 + 4 \times 0.5 + 2$$

$$f(0.5) = 4$$

Q.12 The two sides of a parallelogram are given by the vectors $A = 2\hat{i} - 3\hat{j}$ and $B = 3\hat{i} + 2\hat{j}$. The area of the parallelogram is

- (A) 13 (B) 12 (C) 10 (D) 5

Answer (A)

Solution .

$$A \times B = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -3 & 0 \\ 3 & 2 & 0 \end{vmatrix}$$

$$A \times B = \hat{i}(0 - 0) - \hat{j}(0 - 0) + \hat{k}(4 + 9)$$

$$A \times B = 13 \hat{k}$$

$\therefore$ Area of parallelogram having two side vectors A and B , is given by $= |A \times B|$

$$= |13 \hat{k}|$$

$$= 13.$$

Q.13 In a BOD test, 5 ml of wastewater is diluted with pure water to fill a 300 ml BOD bottle. The initial and final dissolved oxygen contents of the mix are 9.0 mg/l and 7.0 mg/l respectively.

The BOD of wastewater, in mg/l, is

- (A) 2 (B) 10 (C) 120 (D) 600

Answer (C)

Solution .

We have,

$$C_1 \times V_1 = C_2 \times V_2$$

Or Sample BOD $\times$ Sample Volume = BOD Diluted $\times$ Volume Diluted

$$\text{Or BOD Sample} = \frac{\text{BOD Diluted}}{\text{Sample Volume}} \times \text{Volume Diluted}$$

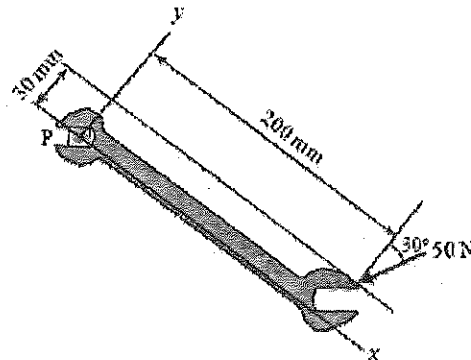
$$\text{Or BOD Sample (mg/l)} = \frac{\text{D.O. Depletion (mg/l)}}{\text{Sample Volume (ml)}} \times \text{Volume Diluted (ml)}$$

[$\because$ D.O. Depletion = D.O. Initial – D.O. final or D.O. 5 Day]

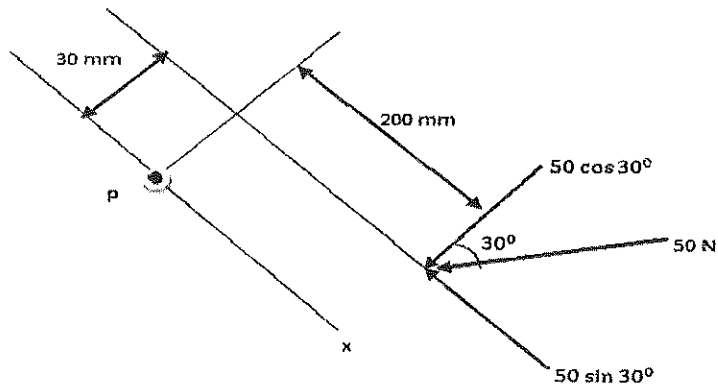
$$\begin{aligned} \therefore \text{ Sample BOD in mg/L} &= \frac{(9 - 7) \times 300}{5} \\ &= 120 \text{ mg/L} \end{aligned}$$

Q.14 A force of 50 N is applied to a wrench as shown in the figure. The magnitude of the moment in N-mm of this force about the point P is _____

Answer : 7900 to 7920.



Solution .



$$\begin{aligned} \text{Magnitude of moment about P} &= 50 \times \cos 30^\circ \times 200 - 50 \times \sin 30^\circ \times 30 \\ &= 7910 \text{ N-mm} \end{aligned}$$

Q.15 Dilatancy of rock is associated with

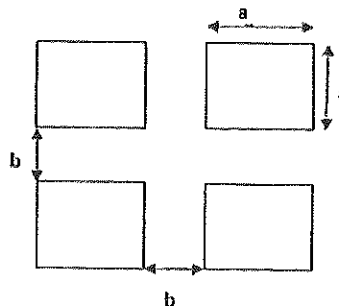
- (A) increase in surface area after fragmentation
- (B) decrease in volume due to compression of rock
- (C) increase in shear strain due to cracking of rock
- (D) increase in volume due to cracking of rock

Answer (D)

Q.16 A bord and pillar panel having square pillars is designed for 30% extraction during development. If the gallery width is 5 m, the side of the pillar in m is _____

Answer : 25 to 26.

Solution .



$$\text{Extraction Ratio} = 1 - \frac{a^2}{(a+b)^2}$$

$$0.30 = 1 - \frac{a^2}{(a+5)^2}$$

$$\text{Or } a^2 = 0.7 (a+5)^2$$

$$\text{Or } a^2 = 0.7 a^2 + 7 a + 17.5$$

$$\text{Or } 0.3 a^2 - 7 a - 17.5 = 0$$

$$\therefore a = 25.6 \text{ \& } -2.28$$

Hence, pillar size is 25.6 m.

Q.17 Low shock and high gas pressure explosive is generally used for the blasting of

- (A) hard and brittle rock mass
- (B) soft and joint rock mass
- (C) hard and massive intact rock mass
- (D) soft and massive intact rock mass

Answer (B)

Q. 18 The covariance of copper grade for a certain lag distance in an ore body is $6.0 (\%)^2$. If the sill is $10 (\%)^2$, the semivariogram for the same lag distance in $(\%)^2$ is

- (A) 4.0
- (B) 16.0
- (C) 2.0
- (D) 64.0

Answer (A)

Solution .

We have,

$$\gamma(h) = C - c(h)$$

Where,

$\gamma(h)$ = semivariogram at lag distance h

C = sill

$c(h)$ = covariance at lag distance h

$$\therefore \gamma(h) = 10 (\%)^2 - 6 (\%)^2$$

$$\therefore \gamma(h) = 4 (\%)^2$$

Q.19 The matrix $A = \begin{bmatrix} -4/6 & 2/6 & 4/6 \\ 4/6 & 4/6 & 2/6 \\ 2/6 & -4/6 & 4/6 \end{bmatrix}$

(A) orthogonal (B) diagonal (C) skew - symmetry (D) symmetric

Answer (A)

Solution .

$$A = \begin{bmatrix} -4/6 & 2/6 & 4/6 \\ 4/6 & 4/6 & 2/6 \\ 2/6 & -4/6 & 4/6 \end{bmatrix}$$

$$\therefore A' = \begin{bmatrix} -4/6 & 4/6 & 2/6 \\ 2/6 & 4/6 & -4/6 \\ 4/6 & 2/6 & 4/6 \end{bmatrix}$$

$$\therefore AA' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore AA' = I$$

Q.20 A gas mixture contains CH_4 , C_2H_6 and H_2 with respective concentrations of 75%, 15% and 10% by volume. The lower explosibility limit of CH_4 , C_2H_6 and H_2 are 5%, 3.3% and 4.2 % respectively. The lower explosibility limit of the gas mixture, in percentage , is _____

Answer : 4.2 to 5.

Solution .

The lower explosibility limit (%) for gas mixture is given by:-

$$X = \left[\frac{100}{\frac{P_1}{N_1} + \frac{P_2}{N_2} + \frac{P_3}{N_3} + \dots + \frac{P_n}{N_n}} \right] \text{ percent}$$

Where,

$P_1, P_2, P_3, \dots$ = contents, in percent by volume, of each combustible component
of the mixture

$N_1, N_2, N_3, \dots$ = lower explosion limit of each components

$$X = \left[\frac{100}{\frac{75}{5} + \frac{15}{3.3} + \frac{10}{4.2}} \right] \%$$

$$X = 4.56 \%$$

- Q.21** Intake air containing 0.2% methane enters a section of an underground mine where emission rate of methane is $0.05 \text{ m}^3/\text{s}$. Assuming that the threshold limit value of methane is 1.25 %, the minimum quantity of fresh air required in m^3/s is _____

Answer : 4.6 to 4.9.

Solution .

We have,

$$Q = \frac{100 q}{C_p - C_a} - q, \text{ m}^3/\text{sec}$$

Where, q = gas emission , m^3/sec

C_a = gas in the intake air, percent

C_p = permissible or safe maximum percent of gas in the return air

$$Q = \frac{100 \times 0.05}{1.25 - 0.2}$$

$$Q = 4.7 \text{ m}^3/\text{s}$$

- Q.22** In a fully mechanized bord and pillar mining system, winning of coal and its transportation from the face is commonly carried out with the combination of

- (A) Continuous miner ,shuttle car, feeder breaker and belt conveyor
- (B) Continuous miner, LHD, feeder breaker and chain conveyor
- (C) Continuous miner , SDL, feeder breaker and belt conveyor
- (D) Continuous miner, shuttle car, feeder breaker and chain conveyor

Answer (A)

- Q.23** An underground coal mine employing 1200 persons experiences 2 fatal injuries, 6 serious injuries and 8 reportable injuries during the year 2013. The total injury rate per 1000 persons employed for the year is _____

Answer : 13.0 to 13.6

Solution .

$$\text{Total injury rate per 1000 persons employed} = \frac{2 + 6 + 8}{1200} \times 1000 = 13.4$$

- Q.24** In self-contained chemical- oxygen self rescuer, oxygen is produced by

- | | |
|----------------------|------------------------|
| (A) Hopcalite | (B) potassium peroxide |
| (C) sodium hydroxide | (D) Protosorb |

Answer (B)

- Q. 25** The failure data of an equipment follows an exponential distribution. If the mean time between failures is 3000 hours, the reliability of the equipment for 750 hours is _____

Answer : 0.75 to 0.81.

Solution .

The reliability that the equipment run up to time t for exponential distribution is given by –

$$R(t) = e^{\frac{-t}{m}} = e^{-\lambda.t}$$

Where,

$$m = \frac{1}{\lambda} = \text{mean time between failure}$$

$$\text{Or } \lambda = \frac{1}{m} = \frac{1}{3000} \text{ per hours}$$

$$\text{And } t = 750 \text{ hours}$$

$$\therefore R(t) = e^{\frac{-750}{3000}} = 0.78$$

Q.26 In a 4.2 m wide and 3.0 m high gallery in a coal seam, twelve shot holes are blaster per round.

The holes are charged with 2 explosive cartridge of 435 g each. If the powder factor of the blast is 2.2 tonne/kg and specific gravity of coal is 1.4, the pull per round of blast in m is

- (A) 1.45 (B) 1.70 (C) 1.30 (D) 4.06

Answer (C)

Solution .

we have,

$$\text{Powder factor} = \frac{\text{coal produced (tonne)}}{\text{explosive required (kg)}}$$

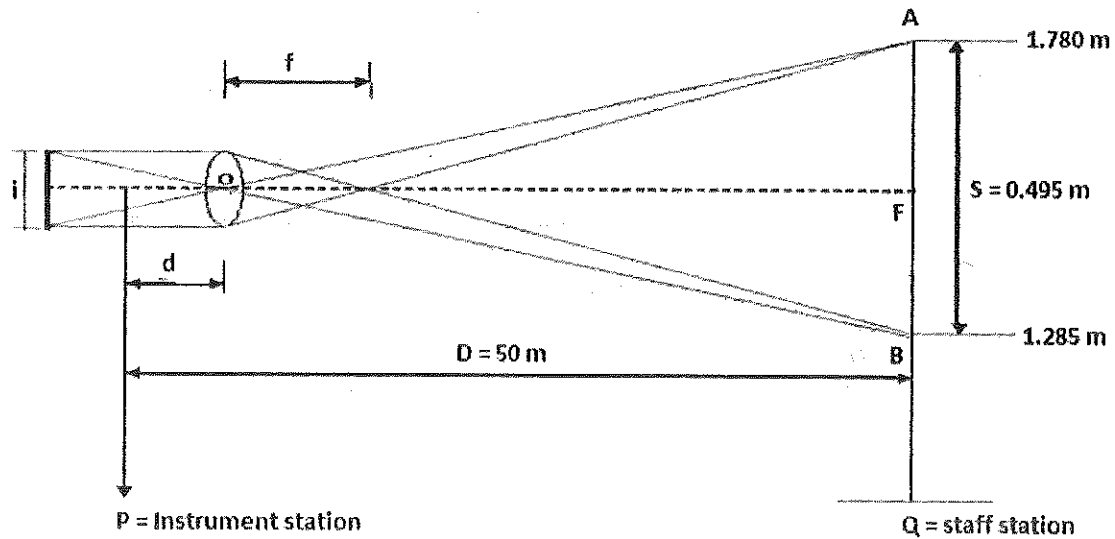
$$\therefore 2.2 = \frac{4.2 \times 3 \times \text{pull} \times 1.4 \text{ per round}}{12 \times 2 \times \frac{435}{1000} \text{ per round}}$$

$$\therefore \text{pull} = 1.30 \text{ m}$$

Q.27 The stadia reading with horizontal sight on a vertical staff held at 50 m from a tacheometer are 1.285 and 1.780 m. The focal length of the object glass is 25 cm and the distance between the object glass and the vertical axis of the tacheometer is 15 cm. the stadia interval in mm is _____

Answer: 2.48 to 2.52

Solution .



We know,

$$\text{Multiplying constant (K)} = \frac{f}{i}$$

Additive constant (C) = ($f + d$) and

$$D = k \cdot s + C$$

Where,

f – focal length

i – stadia interval

D – distance between instrument axis and objective

s – staff intercept

$$\therefore D = k \cdot s + f + d$$

$$\therefore 50 = k \times 0.495 + 0.25 + 0.15$$

$$\therefore k = 100.2$$

$$\text{Now, } k = \frac{f}{i}$$

$$\therefore i = \frac{0.25}{100} = 0.00249 \text{ m} = 2.5 \text{ mm}$$

Q. 28 In a shortwall panel, coal is extracted from the face by a continuous miner having rate of production 30 tonne/hour. Coal having specific gravity of 1.4 is transported by shuttle cars of capacity 0.9 m^3 each to a feeder breaker located at 60 m from the face. If the average speed of the LHD is 0.5 m/s. and total loading and unloading time of LHD is 40 s, the number of LHDs required to match the production of the continuous miner is

- (A) 1 (B) 2 (C) 3 (D) 4

Answer (B)

Solution .

$$\begin{aligned}\text{Continuous miner production} &= 30 \text{ t/h} = \frac{30}{1.4 \times 3600} \text{ m}^3/\text{sec} \\ &= 0.00595 \text{ m}^3/\text{sec}\end{aligned}$$

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

$$0.5 = \frac{60}{\text{time}}$$

$$\therefore \text{Travel time} = 120 \text{ sec and}$$

$$\therefore \text{Cycle time of LHD} = 2 \times 120 + 40 = 280 \text{ sec}$$

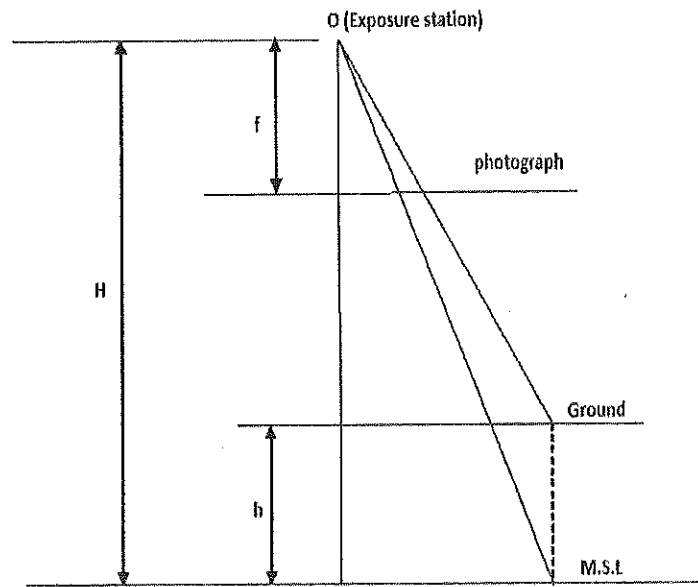
$$\begin{aligned}\text{Coal transported by LHD in one cycle time} &= 0.00595 \times 280 \\ &= 1.67 \text{ m}^3\end{aligned}$$

$$\therefore \text{No. of LHD} = \frac{1.67}{0.9} = 1.9 = 2$$

Q.29 Vertical photographs of an area lying 500 m above the mean sea level are to be taken at a scale of 1:20000 from an aircraft. If the camera has a focal length of 210 mm, the flying height of the aircraft above the mean sea level in m is _____

Answer : 4700.

Solution .



We have ,

$$\text{Scale} = \frac{\text{map distance}}{\text{ground distance}}$$

$$S = \frac{f}{H-h}$$

$$\frac{1}{20000} = \frac{0.210}{H-500}$$

$$\therefore H = 4700 \text{ m}$$

Q .30

Match the following locations with support types in coal mines.

Location	Support type
P. Roadway junctions	1. powered support
Q. Between adjacent panels	2. Chock and bolt
R. Longwall face	3. Back fill
S. Goaf	4. Barrier pillar
(A) P-2,Q-3,R-1,S-4	(B) P-4,Q-3,R-1,S-2

(C) P-2,Q-4,R-1,S-3

(D) P-2,Q-3,R-4,S-1

Answer (C)

Q.31 The value of $\int_0^4 \sqrt{16-x^2} dx$ is

(A) 12.57

(B) 50.24

(C) 25.12

(D) 3.14

Answer (A)

Solution .

We have,

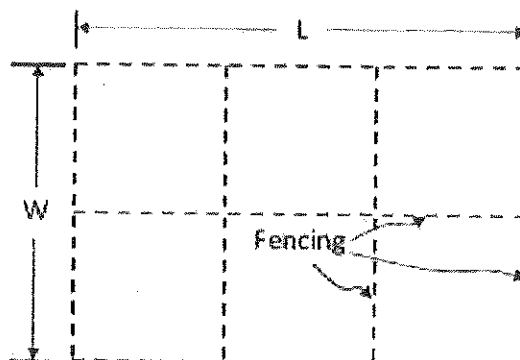
$$\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a}$$

$$\therefore \int \sqrt{4^2 - x^2} dx = \frac{x}{2} \sqrt{4^2 - x^2} + \frac{4^2}{2} \sin^{-1} \frac{x}{4}$$

$$\therefore \int_0^4 \sqrt{4^2 - x^2} dx = \left[\frac{x}{2} \sqrt{4^2 - x^2} + \frac{4^2}{2} \sin^{-1} \frac{x}{4} \right]_0^4$$

$$\therefore \int_0^4 \sqrt{4^2 - x^2} dx = 12.56$$

Q.32 A rectangular field of area $20000 m^2$ is to be divided into 6 different plots by fencing as shown in the figure. The value of L in m for which the total length of fencing becomes minimum is _____



Answer : 161 to 165.

Solution .

Let total fencing length , $Z = 4W + 3L$

∴ Area of rectangular = $W \times L = 20000 \text{ m}^2$

$$\therefore Z = 4 \times \frac{20000}{L} + 3L \dots\dots\dots(1)$$

Differentiating (1) w. r. t to L

$$\frac{dZ}{dL} = -\frac{80000}{L^2} + 3$$

For minimum value,

$$\frac{dZ}{dL} = 0$$

Hence,

$$-\frac{80000}{L^2} = -3$$

$$\therefore L = 163.3 \text{ m}$$

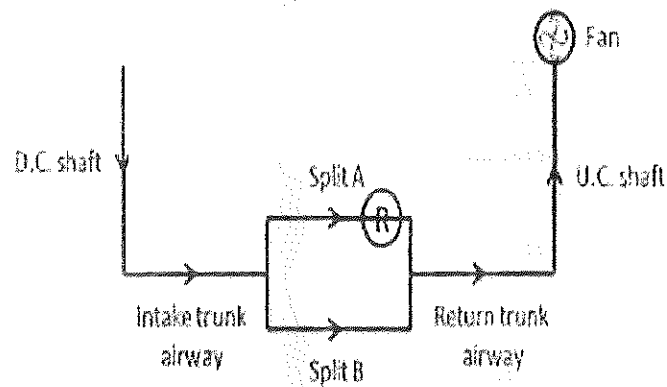
Q.33 Match the following for a drilling system

Component	Function
P. Drill	1. Utilization of energy in fragmenting rock
Q. Drill rod	2. Reduction of energy loss due to regrinding
R. Drill bit	3. Conversion of original form of energy in to mechanical energy
S. Flushing medium	4. Transmission of energy from prime mover to applicator
(A) P-3,Q-1,R-2,S-4; (B) P-4,Q-1,R-3,S-2; (C) P-3,Q-4,R-1,S-2; (D) P-2,Q-1,R-3,S-4;	

Answer (C)

Q.34 For the ventilation system shown, the combined resistance of the trunk airways and the shaft is $2.2 \text{ N s}^2 \text{ m}^{-8}$. The resistance of the split A and B are $0.5 \text{ N s}^2 \text{ m}^{-8}$ and $0.8 \text{ N s}^2 \text{ m}^{-8}$ respectively. A regulator of size 2.0 m^2 is placed in split A. Considering the fan generates a pressure of 1000 Pa , the air flow in m^3/s in split B is _____

Answer : 10.2 to 10.8.



Solution .

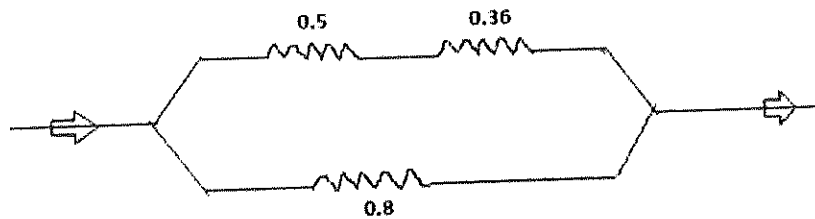
We have,

$$\text{Area of regulator} = \frac{1.2}{\sqrt{R}}, \text{ m}^2$$

Where, R - resistance of regulator

$$\therefore 2 = \frac{1.2}{\sqrt{R}}$$

$$\therefore R = 0.36 \text{ N s}^2 \text{ m}^{-8}$$



$$\frac{1}{\sqrt{R_1}} = \frac{1}{\sqrt{0.5 + 0.36}} + \frac{1}{\sqrt{0.8}}$$

$$\therefore R_1 = 0.2 \text{ } Ns^2m^{-8}$$

$$\text{Total resistance of mine} = 2.2 + 0.2 = 2.4 \text{ } Ns^2m^{-8}$$

$$\therefore P = R_T Q_T^2$$

$$\therefore 1000 = 2.4 \times Q_T^2$$

$$\therefore Q_T = 20.4 \text{ } m^3/sec$$

Here, Pressure drop = pressure drop in trunk roadway + pressure drop in split A

$$1000 = 2.2 \times (20.4)^2 + 0.86 \times Q_A^2$$

$$\therefore Q_A = 9.9 \text{ } m^3/sec \text{ and}$$

$$\text{Now, } Q_T = Q_A + Q_B$$

$$20.3 = 9.9 + Q_B$$

$$\therefore Q_B = 10.5 \text{ } m^3/sec$$

Q.35 A mine fan running at 300 rpm delivers $150 \text{ } m^3/s$ of air at a pressure of 900 Pa. Fan and motor efficiencies are 75% and 90% respectively. If the fan speed is reduced to 250 rpm, the saving in electric power input to motor in kW is _____

Answer : 82 to 86.

Solution .

$$\text{Actual power developed } (P_{ac}) = PQ \times 10^{-3} kW$$

$$P_{ac} = 900 \times 150 \times 10^{-3} = 135 \text{ kW}$$

The relation between power (P) and fan speed (N) is given by-

$$P \propto N^3$$

$$\therefore \frac{(P_{ac})_1}{(P_{ac})_2} = \left(\frac{N_1}{N_2}\right)^3$$

$$\frac{135}{(P_{ac})_2} = \left(\frac{300}{250}\right)^3$$

$$\therefore (P_{ac})_2 = 78.12 \text{ kW}$$

$$\therefore \text{Input power to the motor} = \frac{\text{output power}}{\text{efficiency}}$$

Therefore ,

$$\begin{aligned} \text{Saving in electric power input to the motor} &= \frac{135 - 78.12}{0.75 \times 0.90}, \text{ kW} \\ &= 84.27 \text{ kW} \end{aligned}$$

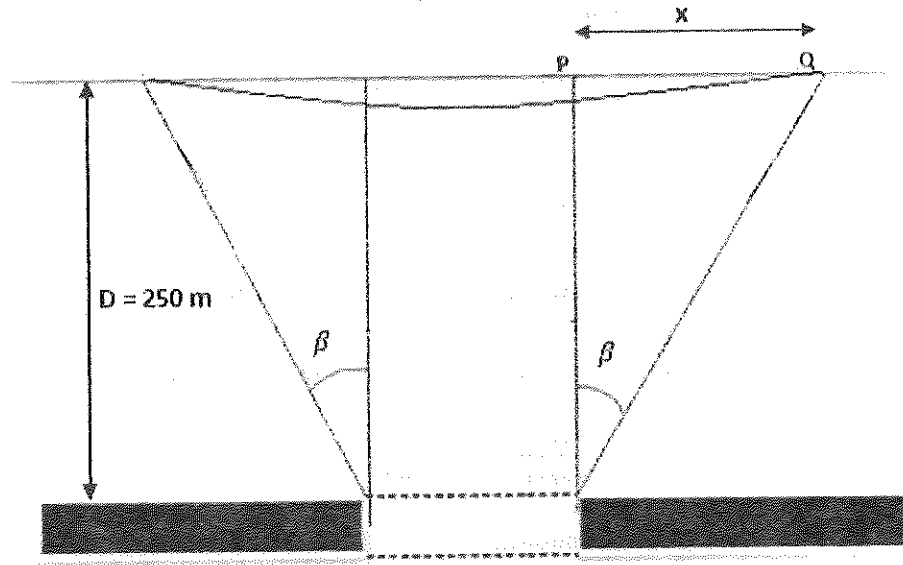
Q.36 Subsidence profile function , $s(x)$, along the lateral cross- section over a flat longwall panel is given as

$$s(x) = 0.8 \left[0.996 - \tanh \left(\frac{8.3x}{D} \right) \right], m$$

Where x = distance (m) from the inflection point and D = depth (m) of seam. Considering that the inflection point lies vertically above the edge of the panel, the angle of draw in degrees for a depth of 250 m is _____

Answer : 20 to 21.

Solution .



Where,

P - point of inflection

Q - point of zero subsidence

$$s(x) = 0.8 \left[0.996 - \tanh \left(\frac{8.3x}{D} \right) \right], \text{ m}$$

At point Q , $s(x) = 0$; then

$$0 = 0.8 \left[0.996 - \tanh \left(\frac{8.3x}{250} \right) \right]$$

$$0.0332x = \tanh^{-1}(0.996)$$

$$0.0332x = \frac{1}{2} \ln \left(\frac{1+0.996}{1-0.996} \right)$$

$$\therefore x = 93.56 \text{ m}$$

Therefore,

$$\text{Angle of draw } (\beta) = \tan^{-1} \left(\frac{93.26}{250} \right)$$

$$\therefore \beta = 20.5^\circ$$

Q. 37 A goaf void of 250 m^3 is filled in 3 hours by hydraulic sand stowing method. Density of the

sand is 2.6 tonne/m^3 . If the filling factor of goaf void is 0.9 and sand to water ratio in the stowing mixture is 1.0 tonne to 1.1 m^3 , the stowing rate in m^3/h is _____

Answer : 286 to 293.

Solution .

$$\text{Volume of sand required} = \frac{250 \times 0.9}{3} = 75 \text{ m}^3/\text{hour}$$

$$\text{Mass flow rate of sand} = 75 \times 2.6 = 195 \text{ tonne /hour}$$

$$\therefore \frac{\text{sand}}{\text{water}} = \frac{1 \text{ tonne}}{1.1 \text{ m}^3}$$

$$\text{Volume of water required} = 195 \times 1.1 = 214.5 \text{ m}^3/\text{hour}$$

Hence,

$$\text{Stowing rate} = \text{volume of (water + sand)required ,m}^3/\text{hour}$$

$$= 75 + 214.5, \text{m}^3/\text{hour}$$

$$= 289.5 \text{ m}^3/\text{hour}$$

Q.38 A single-acting reciprocating pump delivers $0.018 \text{ m}^3/\text{s}$ of water when running at 45 cycles per minute. The piston diameter is 300 mm and stroke length is 400 mm. The volumetric efficiency of the pump in % is _____

Answer : 83 to 87.

Solution .

$$\text{Volumetric displacement} = \text{piston area} \times \text{stroke length}$$

$$= \frac{\pi}{4} \times 0.3^2 \times 0.4, \text{m}^3$$

$$= 0.028 \text{ m}^3$$

$$\text{Theoretical discharge} = \text{volumetric displacement} \times \text{rpm}$$

$$= 0.028 \times 45 \text{ m}^3/\text{min}$$

$$= 0.021 \text{ m}^3/\text{sec}$$

$$\therefore \text{volumetric efficiency } (\eta_v) = \frac{\text{Actual flow}}{\text{Theoretical flow}} \times 100$$

$$= \frac{0.018}{0.021} \times 100$$

$$\eta_v = 86 \%$$

Q.39 Match the method of mining with strength of ore body, type of support and ore body geometry

Strength	Support	Geometry	Method
P. Strong	L. Unsupported	X. Tabular and steep	1. Cut-and-fill
Q. Moderate	M. Artificially supported	Y. Tabular and flat	2. Block caving
R. Weak	N. Self supporting	Z. Massive and steep	3. Room and pillar

(A) P-M-X-3, Q-N-Z-2, R-L-Y-1

(B) P-L-X-1, Q-N-Z-3, R-M-Y-2

(C) P-N-Y-3, Q-M-X-1, R-L-Z-2

(D) P-L-Z-1, Q-N-Y-3, R-M-X-2

Answer (C)

Q.40 A mine air sample contains CH_4 , CO , H_2 , N_2 and O_2 . The mine air analysis using Haldane apparatus gives the following results expressed in percentage of total sample volume.

Total contraction after combustion : 10.0

CO_2 formed after combustion : 6.0

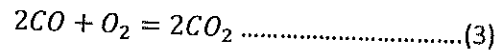
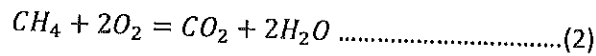
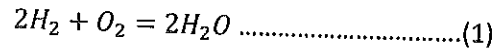
O_2 consumed in combustion : 9.5

The percentage of CH_4 in the sample analysed is _____

Answer : 3.8 to 4.2.

Solution .

Hydrogen, methane and carbon dioxide react with oxygen as follows :-



From equation (1), (2) & (3), the following relations are obtained :-

$$\text{Contraction} = \frac{3}{2}H_2 + 2CH_4 - \frac{1}{2}CO \dots\dots\dots(4)$$

$$\text{Carbon dioxide} = CH_4 + CO \dots\dots\dots(5)$$

$$\text{Oxygen consumed} = \frac{1}{2}H_2 + 2CH_4 + \frac{1}{2}CO \dots\dots\dots(6)$$

From equation (4), (5) & (6)

$$H_2 = \text{Contraction} - \text{oxygen consumed} \dots\dots\dots(7)$$

$$CH_4 = \frac{2 \times \text{contraction} - CO_2}{3} - H_2 \dots\dots\dots(8)$$

$$\therefore H_2 = 10 - 9.5 = 0.5\%$$

$$\therefore CH_4 = \frac{2 \times 10 - 6}{3} - 0.5 = 4.2 \%$$

Q.41 The initial investment for a small scale mining project is Rs. 5.0 crore. Annual cash inflow for a life period of 4 years is given below.

Year	Cash inflow (Rs. crore)
1	1.5
2	2.0
3	2.0
4	1.5

The net present value of the project at an annual discount rate of 10 % in Rs. crore is _____

Answer : 0.5 to 0.6.

Solution .

We have,

$$\text{Net present value} = -X + \frac{R_1}{(1+i)} + \frac{R_2}{(1+i)^2} + \dots + \frac{R_n}{(1+i)^n}$$

$$\text{NPV} = -5 + \frac{1.5}{(1+0.1)} + \frac{2}{(1+0.1)^2} + \frac{2}{(1+0.1)^3} + \frac{1.5}{(1+0.1)^4}$$

$$\therefore \text{NPV} = 0.6 \text{ Crore rupees}$$

Q.42 Given the following linear programming problem,

$$\text{Maximize } z = 3x_1 + 4x_2$$

$$\text{Subject to } 2x_1 + x_2 \leq 6$$

$$2x_1 + 3x_2 \leq 9$$

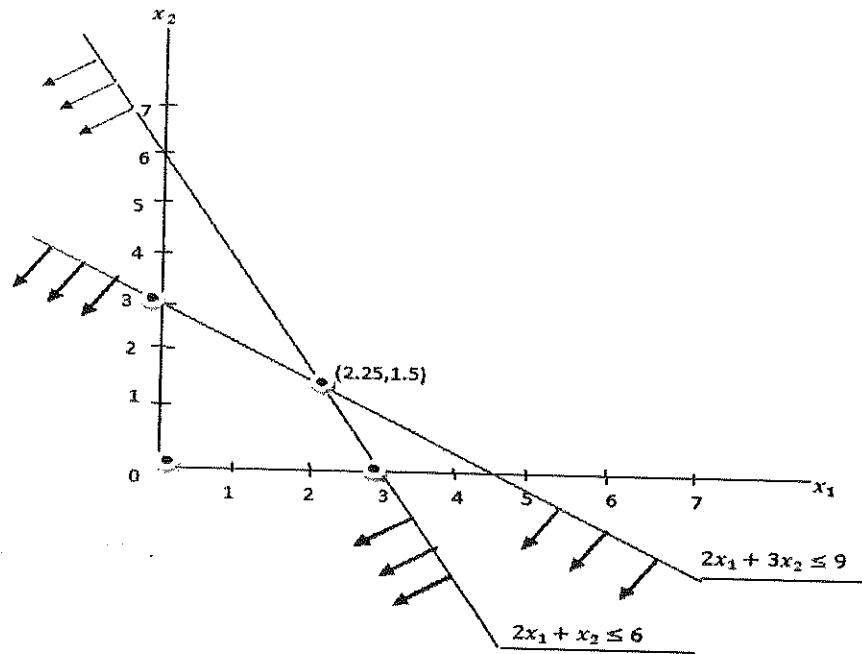
$$x_1 \geq 0, x_2 \geq 0$$

The corner point feasible solution in terms of (x_1, x_2) is

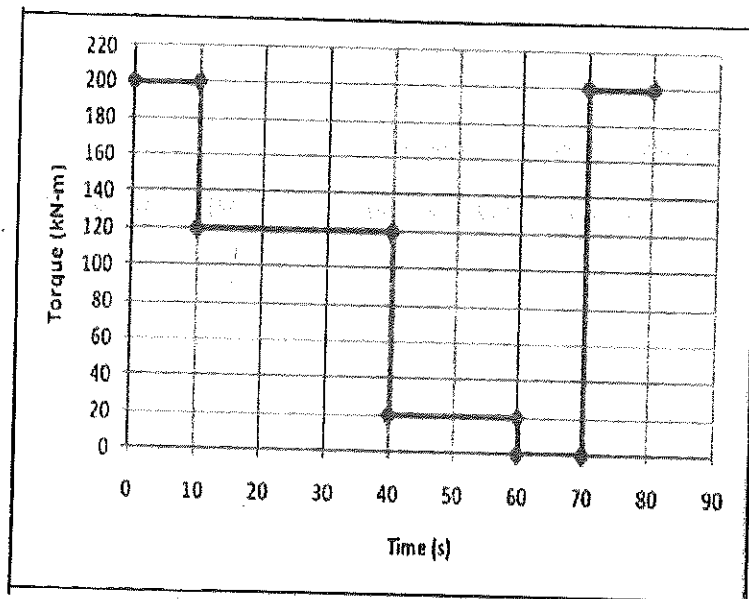
(A) (1.5, 0) (B) (1.25, 1.5) (C) (0.5, 1.0) (D) (2.25, 1.5)

Answer (D)

Solution .



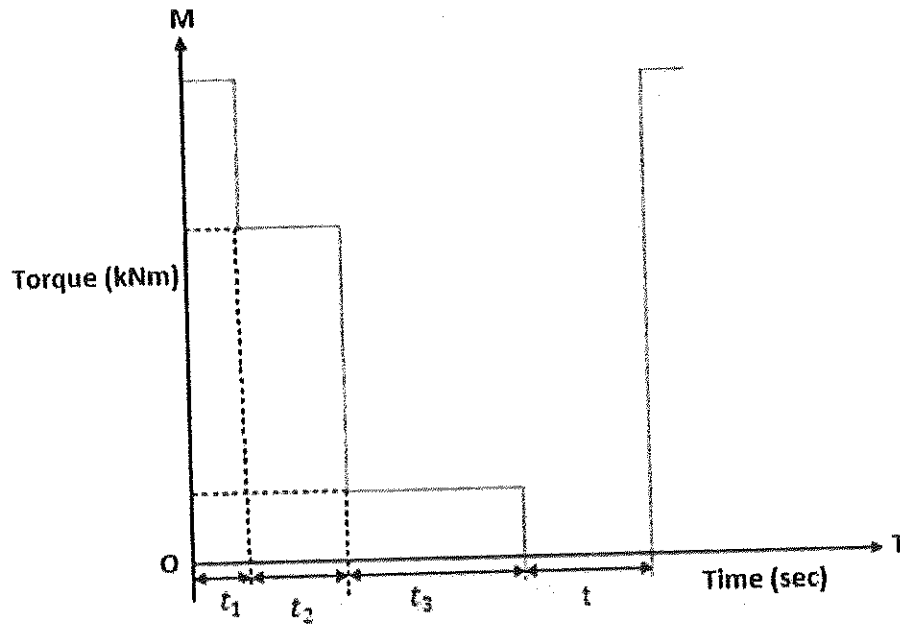
Q. 43 The 3 period torque – time diagram of a statically balanced hoist is shown in the figure.



The rms torque for the motor in kN-m is _____

Answer : 106 and 113 .

Solution .



$$\text{r.m.s torque, } M_{rms} = \sqrt{\frac{\sum_0^{n_i} \int_0^{t_i} M^2 dt}{T+t}}$$

where,

$$T = t_1 + t_2 + t_3 = \text{running time and } t = \text{rest time}$$

$$\text{r.m.s torque, } M_{rms} = \sqrt{\frac{200^2 \times 10 + 120^2 \times 30 + 20^2 \times 20}{10 + 30 + 20 + 10}}$$

$$\therefore M_{rms} = 109.55 \text{ kN.m}$$

Q.44 Airborne PM_{10} concentration in a residential area is monitored for 24 hours by a respirable dust sampler. Initial and final weights of the filter paper are 2.3125 g and 2.6996 g respectively. The average airflow rate during sampling is $1.2 \text{ m}^3/\text{min}$. The PM_{10} concentration of the area in μgm^{-3} is _____

Answer : 220 and 228.

Solution .

Total suspended solids/suspended particulate matter is given by-

$$\text{TSS/SPM} = \frac{(A - B)}{C}, g/m^3$$

Where,

A - weight of filter plus retained solids in $g = 2.6996 \text{ gm}$

B - weight of clean filter in $g = 2.3125 \text{ gm}$

C - volume of sample filtered in m^3

Given that $C = 1.2 \text{ m}^3/\text{minute}$

$$\therefore 24 \text{ hours} = 24 \times 1.4 \times 60 \text{ m}^3$$

Therefore,

$$\text{TSS/SPM} = \frac{2.6996 - 2.3125}{1.2 \times 24 \times 60} = 2.24 \times 10^{-4} g/m^3$$

$$\text{TSS/SPM} = 224 \mu g/m^3$$

Q.45 The assignment problem given requires four different jobs to be done on four different machines.

Job	Machine			
	M_1	M_2	M_3	M_4
J_1	27	35	36	39
J_2	33	37	36	35
J_3	30	26	28	24
J_4	38	29	35	33

The minimum cost of assignment is _____

Answer : 116.

Solution .

Step 01. Subtract the smallest element of each row from all the element of the row . So there will be at least one zero in each row.

0	8	9	3
0	4	3	2
6	2	4	0
9	0	6	4

Step 02. Subtract the minimum element of the column not containing zero from all the elements of that column.

0	8	6	3
0	4	0	2
6	2	1	0
9	0	3	4

Step 03. Draw minimum number of lines to cover all the zeros in the table. If number of lines drawn equal to the order of the matrix, then optimum solution is obtained. If not, then subtract the smallest uncovered element from all the uncovered elements. Add it to the elements that lie at the intersection of lines.

0	8	6	3
0	4	0	2
6	2	1	0
9	0	3	4

Step 04. Examine rows successively until a row with exactly one unmarked zero is found. Mark ($\square$) this zero. Mark ($\times$) all other zeros in the same column .

$\square$ 0	8	6	3
0	4	$\square$ 0	2
6	2	1	$\square$ 0
9	$\square$ 0	3	4

This is feasible solution.

Hence,

$$\begin{aligned}\text{Minimum cost of assignment} &= 27 + 36 + 24 + 29 \\ &= 116\end{aligned}$$

Q. 46 Acceleration of a particle moving in a straight line is expressed by

$$\frac{d^2s}{dt^2} = 2t$$

Where, s denotes distance (m) and t , time (s). At time $t=0$, the distance and velocity of the particle are 0 m and 3 m/s respectively. The distance travelled by the particle in m after 3 s is

- (A) 3 (B) 6 (C) 9 (D) 18

Answer (D)

Solution .

$$\frac{d^2s}{dt^2} = 2t$$

Or $a = 2t$

$$\text{Or } \frac{dv}{dt} = 2t$$

$$\text{Now, } \int dv = \int 2t \, dt$$

$$V = t^2 + c \dots\dots\dots(1)$$

$$\text{at } t = 0$$

$$V = 3$$

$$\therefore 3 = 0 + c$$

$$\therefore C = 3$$

Equation (1) becomes,

$$V = t^2 + 3 \dots\dots\dots(2)$$

$$\text{Or } \frac{ds}{dt} = t^2 + 3$$

$$\int ds = \int (t^2 + 3) \, dt$$

$$s = \frac{t^3}{3} + 3t + k \dots\dots\dots(3)$$

$$\text{at } t = 0$$

$$s = 0$$

$$\therefore 0 = 0 + 0 + k$$

$$K = 0$$

Equation (3) becomes

$$S = \frac{t^3}{3} + 3t$$

$$\text{at } t = 3 \text{ sec}$$

$$s = 18 \text{ m}$$

Q .47 Rock bolt have length $L = (150 + X)$ cm, where X is a random variable with probability density function

$$f(x) = \frac{1}{4} (1 - 3x), \text{ if } -2 \leq x \leq 2$$

0, otherwise

If 95% bolt length (L) lie in the interval $150 - c$ to $150 + c$ cm, the value of c is _____

Answer : 1.88 to 1.92.

Solution .

Probability density function of random variable X is defined by-

$$\int_{-\infty}^{+\infty} f(x) dx = 1$$

$$\text{Here, } \int_{-c}^{+c} \frac{1}{4} (1 - 3x) dx = 0.95$$

$$\left[\frac{1}{4}x - \frac{1}{4} \cdot \frac{3x^2}{2} \right]_{-c}^{+c} = 0.95$$

$$\frac{1}{4}c - \frac{1}{4} \cdot \frac{3c^2}{2} + \frac{1}{4}c + \frac{1}{4} \cdot \frac{3c^2}{2} = 0.95$$

$$\frac{1}{2}c = 0.95$$

$$\therefore c = 1.90$$

Q. 48 The properties for a bivariate distribution of two random variables X and Y are given below.

$$E(X) = 24, E(Y) = 36, E(X^2) = 702, E(Y^2) = 1524, E(XY) = 1004$$

The correlation coefficient between X and Y is _____

Answer : 0.8 and 0.85

Solution .

The correlation coefficient of the bivariate distributed random variable X and Y is-

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_x \cdot \sigma_y} \dots\dots\dots(1)$$

Where,

$\text{Cov}(X,Y)$ = covariance between bivariate distributed random variables X and Y

$$\text{Corr}(X,Y) = E(XY) - \mu_x \mu_y$$

$$E(X) = \mu_x = \text{mean value of } X$$

$$E(Y) = \mu_y = \text{mean value of } Y, \text{ and}$$

$$\text{Variance of } X = \sigma_x^2 = E(X^2) - [E(X)]^2$$

$$\text{Variance of } Y = \sigma_y^2 = E(Y^2) - [E(Y)]^2$$

Therefore,

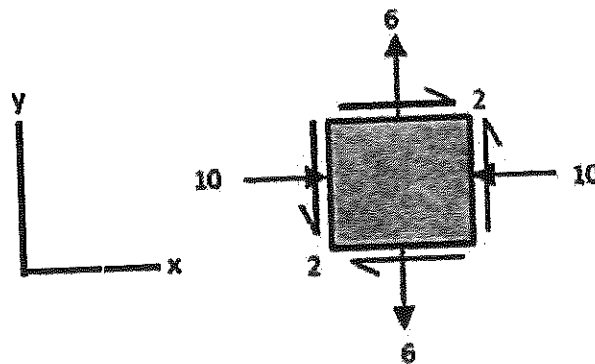
From equation (1)

$$\text{Corr}(X,Y) = \frac{E(XY) - \mu_x \mu_y}{\sqrt{E(X^2) - [E(X)]^2} \sqrt{E(Y^2) - [E(Y)]^2}}$$

$$\text{Corr}(X,Y) = \frac{1004 - 24 \times 36}{\sqrt{702 - 24^2} \sqrt{1524 - 36^2}}$$

$$\text{Corr}(X,Y) = 0.83$$

Q. 49 Biaxial stresses at a point inside a pillar are shown in the figure.



All stress values are in Mpa

The magnitude of the maximum shear stress in MPa and its direction with the x-axis in degrees at the same point respectively are

- (A) 8.25, 37.98 (B) 7.49, 37.98 (C) 8.25, 52.02 (D) 7.49, 52.02

Answer (A)

Solution .

$$\text{Here, } \sigma_{xx} = -10 \text{ MPa}$$

$$\sigma_{yy} = 6 \text{ MPa}$$

$$\tau_{xy} = 2 \text{ MPa}$$

Principal stresses :- considering $\sigma_1 > \sigma_3$

$$\sigma_{1,3} = \left(\frac{\sigma_{xx} + \sigma_{yy}}{2} \right) \pm \sqrt{\frac{1}{4}(\sigma_{xx} - \sigma_{yy})^2 + \tau_{xy}^2}$$

$$\sigma_{1,3} = \left(\frac{-10 + 6}{2} \right) \pm \sqrt{\frac{1}{4}(-10 - 6)^2 + 2^2}$$

$$\sigma_{1,3} = -2 \pm 8.25$$

$$\sigma_1 = 6.25 \text{ MPa } (\because \text{taking +ve}) \text{ and}$$

$$\sigma_3 = -10 \text{ MPa } (\because \text{taking -ve})$$

$$\therefore \text{Maximum shear stress } (\tau_{max}) = \frac{\sigma_1 - \sigma_3}{2} = 8.25 \text{ MPa, and}$$

Direction of shear stress with x- axis is given by:-

$$\tan 2\theta_s = - \frac{(\sigma_{xx} - \sigma_{yy})}{2\tau_{xy}}$$

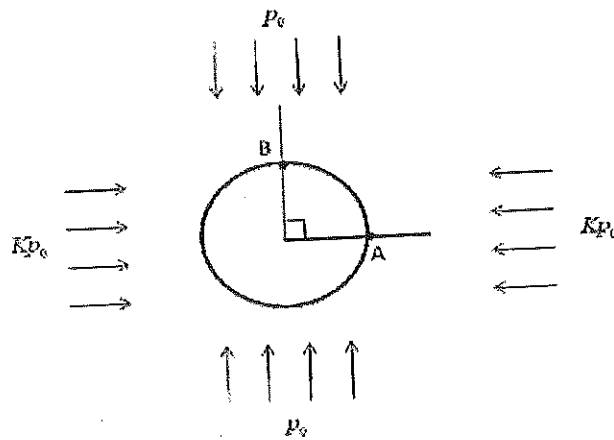
$$\tan 2\theta_s = - \frac{(-10 - 6)}{2 \times 2}$$

$$\tan 2\theta_s = 4$$

$$\therefore \theta_s = 37.98^\circ$$

Q,50

A circular tunnel is constructed in a biaxial far field stress (vertical stress p_0 and horizontal stress Kp_0) as shown in the figure.

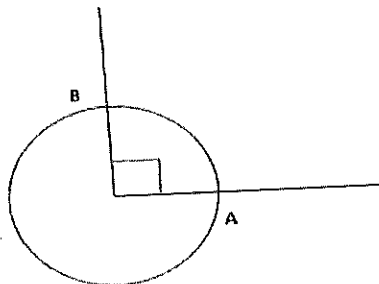
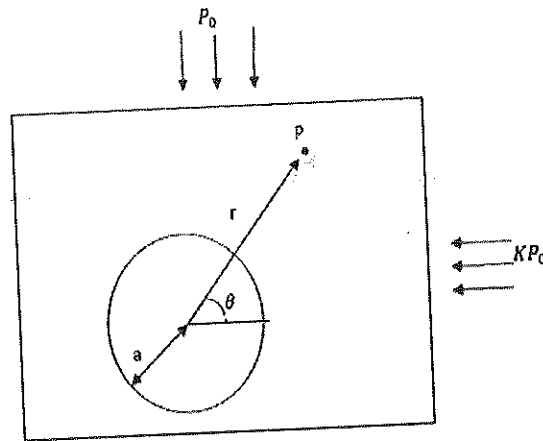


If the ratio of the tangential stress measured at the boundary points A and B is 3:1, the value of

K is _____

Answer : 0.6.

Solution .



Tangential stress(σ_θ) is given by:

$$\sigma_{\theta\theta} = \frac{1}{2} P_z \left\{ (1 + K) \left(1 + \frac{a^2}{r^2} \right) + (1 - K) \left(1 + 3 \frac{a^4}{r^4} \right) \cos 2\theta \right\} \dots\dots\dots(1)$$

But at boundary $a = r$ then, equation (1) becomes

$$\sigma_{\theta\theta} = P[(1 + k) + 2(1 - k) \cos 2\theta] \dots\dots\dots(2)$$

At point A, equation (2) becomes ,putting $\theta = 0^\circ$

$$(\sigma_{\theta\theta})_A = P[(1 + k) + 2(1 - k)] \dots\dots\dots(3)$$

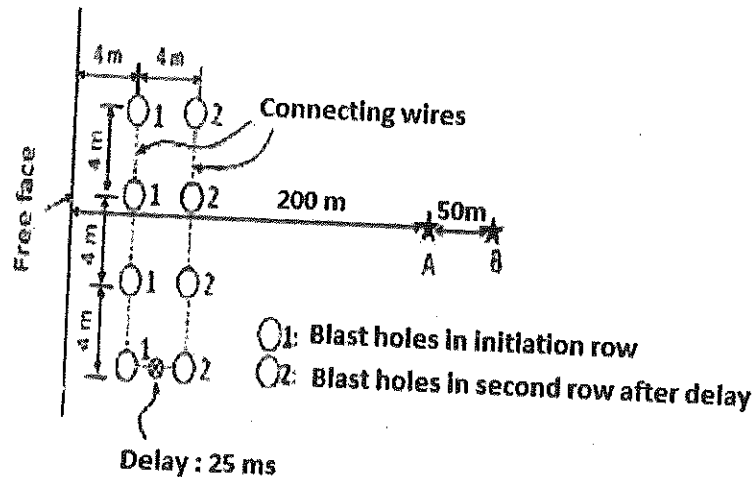
At point B equation (2) becomes, putting $\theta = 90^\circ$

$$(\sigma_{\theta\theta})_B = P[(1 + k) - 2(1 - k)] \dots\dots\dots(4)$$

$$\therefore \frac{(\sigma_{\theta\theta})_A}{(\sigma_{\theta\theta})_B} = \frac{3}{1} = \frac{P[(1 + k) + 2(1 - k)]}{P[(1 + k) - 2(1 - k)]}$$

$$\therefore k = 0.6$$

Q.51 Peak particle velocity (PPV) at points A and B are measured for a blast pattern as shown in the figure.



The relevant data are :

Amount of explosive per hole in the 1st row : 500 kg

Amount of explosive per hole in the 2nd row : 475 kg

PPV at point A : 18 mm/s

PPV at point B

: 10 mm/s

Considering the following relationship,

$$PPV = K \left(\frac{D}{\sqrt{Q}} \right)^{-n}, \text{mm/s}$$

Where D (in m) denotes the distance from the blast row to the measuring point and Q (in kg), maximum charge per delay. The site constant K and n respectively are

- (A) 1002, 3.13 (B) 622, 2.92 (C) 823, 2.59 (D) 1245, 2.99

Answer (C)

Solution .

We have,

$$PPV = K \left(\frac{D}{\sqrt{Q}} \right)^{-n}, \text{mm/sec}$$

At point A,

$$18 = K \left(\frac{196}{\sqrt{4 \times 500}} \right)^{-n} \dots\dots\dots(1)$$

At point B,

$$10 = K \left(\frac{246}{\sqrt{4 \times 500}} \right)^{-n} \dots\dots\dots(2)$$

From equation (1) & (2)

$$1.8 = \left(\frac{196}{246} \right)^{-n}$$

$$\text{Or } 1.8 = \left(\frac{246}{196} \right)^n$$

Taking logarithm both sides, then

$$\ln(1.8) = n \ln \left(\frac{246}{196} \right)$$

$$\therefore n = 2.5868$$

Putting n = 2.59, in equation (1), then

$$\therefore K = 823$$

Q.52 Copper ore of average grade 0.65% is mined, milled, smelted and then refined. The following information is available:

Mill recovery rate	: 85%
Average grade in mill concentrate	: 20%
Loss in smelting process	: 5 kg/tonne of concentrate
Loss in refining process	: 2 kg/tonne of blister copper

The amount of refined copper obtained per tonne of ore in kg is

- (A) 5.10 (B) 5.37 (C) 5.52 (D) 6.50

Answer (B)

Solution .

Grade of ore = 0.65 %

- a) Copper content per tonne of ore = $\frac{0.65}{100} \times 1000 = 6.5 \text{ kg}$
 b) Copper recovered by mill = $6.5 \times \frac{85}{100} = 5.52 \text{ kg}$
 c) Concentration ratio (r) = $\frac{\text{amount of copper per tonne of concentrate}}{\text{amount of copper per tonne of ore}}$

Now,

100 kg of concentrate has = 20 kg of copper

1000 kg of concentrate has = 200 kg of copper

$$\therefore r = \frac{200}{5.52} = 36.23$$

- d) smelting loss per tonne of ore = 5 kg/tonne of concentrate
 $= 5 \text{ kg}/36.17 \text{ tonne of ore}$

$\therefore$ smelting loss = 0.138 kg/tonne of ore

Blister copper produced per tonne of ore = $5.52 - 0.138 = 5.38 \text{ kg}$

e) Refinery loss for 5.38 kg of blister copper = $\frac{2 \times 5.38}{1000} = 0.0107$ kg

Hence,

$$\begin{aligned}\text{Refined copper produced per tonne of ore} &= 5.38 - 0.0107 \\ &= 5.37 \text{ kg}\end{aligned}$$

Q.53 The ratio of horizontal to vertical in-situ stresses, K , at a mine field varies with depth, D (in m) as

$$K = \frac{267}{D} + 1.25$$

If the unit weight of overburden rock is 25 kN/m^3 , the horizontal stress in MPa at a depth of 400 m is _____

Answer : 19.10 and 19.25.

Solution .

$$\text{Vertical stress } (\sigma_v) = \gamma h$$

Where,

γ - unit weight of rock

h – depth below earth surface

$$\sigma_v = 25 \times 400 = 10000 \text{ kN/m}^3, \text{ and}$$

$$K = \frac{267}{D} + 1.25$$

$$K = \frac{267}{400} + 1.25$$

$$K = 1.9175$$

Now, relation between horizontal stress (σ_h) and vertical stress (σ_v) is given by-

$$\sigma_h = k. \sigma_v$$

$$= 1.9175 \times 10000 \text{ kN/m}^3$$

$$\therefore \sigma_h = 19.2 \text{ MPa}$$

- Q.54** A coal seam of 2 m thickness is extracted by a longwall retreating panel with face length of 120 m. Web depth of shearer is 0.6 m. Average manpower in the longwall face in a shift is 20. The specific gravity of in-situ coal is 1.4. If the shearer makes 4 full-face cuts in 3 shifts, the face OMS in tonne is _____

Answer : 13 to 14.

Solution .

$$\text{OMS} = \frac{\text{production in one shift (tonne)}}{\text{numbers of workers in one shift}}$$

In 3 shift = 4 full face cut , then

$$\text{In 1 shift} = \frac{4}{3} \text{ full face cut} = 1.34 \text{ full face cut}$$

$$\text{OMS} = \frac{\text{production in one shift (tonne)}}{\text{numbers of workers in one shift}}$$

$$\text{OMS} = \frac{2 \times 120 \times 0.6 \times 1.4 \times 1.34}{20}$$

$$\text{OMS} = 13.5 \text{ tonnes}$$

Or

$$\text{OMS} = \frac{\text{production in three shifts (tonne)}}{\text{numbers of workers in three shifts}}$$

$$\text{OMS} = \frac{2 \times 120 \times 0.6 \times 1.4 \times 4}{3 \times 20}$$

$$\text{OMS} = 13.5 \text{ tonnes}$$

- Q.55** A loaded dumper of total mass 75 tonne, having wheel diameter 1250 mm, runs on a haul road which offers an average specific rolling resistance of 260 N/tonne. The engine develops

an axle torque of 15 kN-m. The starting acceleration of the dumper in m/s^2 is _____

Answer : 0.055 and 0.065

Solution .

We know,

$$\text{Torque} = \text{Tractive force} \times \text{radius}$$

$$15 \text{ kN.m} = \text{Tractive force} \times \frac{1.25}{2}$$

$$\therefore \text{Tractive force} = 24 \text{ kN}$$

The tractive force exerted by engine is given by :-

$$\text{Tractive force} = \text{accelerating force} + \text{frictional resistance}$$

$$24 = \text{accelerating force} + \frac{75 \times 260}{1000}$$

$$\therefore \text{Accelerating force} = 4.5 \text{ kN} = 4500 \text{ N}$$

Now, $\text{Accelerating force} = \text{Mass} \times \text{acceleration}$

$$4500 = 75000 \times \text{acceleration}$$

$$\therefore \text{Acceleration} = 0.6 \text{ m}^2/\text{sec}$$

Q.56 Choose the appropriate word/phrase ,out of the four options given below, to complete the following sentence:

Apparent lifelessness _____ dormant life.

(A) harbours (B) leads to (C) supports (D) affects

Answer (A)

Q.57 Fill in the blank with correct idiom/phrase.

That boy from the town was a _____ in the sleepy village.

(A) dog out of herd (B) sheep from the heap

(C) fish out of water (D) bird from the flock

Answer (C)

Q.58 Choose the statement where underlined word is used correctly.

(A) When the teacher eludes to different authors, he is being elusive.

(B) When the thief keeps eluding the police, he is being elusive.

(C) Matters that are difficult to understand, identify or remember are allusive.

(D) Mirages can be allusive, but a better way to express them is illusory.

Answer (B)

Q.59 Tanya is older than Eric.

Cliff is older than Tanya.

Eric is older than Cliff.

If the first two statements are true, then the third statement is :

(A) True

(B) False

(C) Uncertain

(D) Data insufficient

Answer (B)

Q.60 Five teams have to complete in a league, with every team playing every other team exactly once, before going to the next round. How many matches will have to be held to complete the league round of matches ?

(A) 20

(B) 10

(C) 8

(D) 5

Answer (B)

Solution .

The number of all combinations of n things, taken r at a time is :-

$$n_{C_r} = \frac{n!}{r!(n-r)!}$$

$$5_{C_2} = \frac{5!}{2!(5-2)!}$$

$$= \frac{5 \times 4}{2}$$

$$5_{C_2} = 10$$

Q.61 Select the appropriate option in place of underlined part of sentence.

Increased productivity necessary reflects greater efforts made by the employees.

- (A) Increases in productivity necessary
- (B) Increase productivity is necessary
- (C) Increase in productivity necessarily
- (D) No improvement required

Answer (C)

Q.62 Given below are two statements followed by two conclusions. Assuming these statements to be true, decide which one logically follows.

Statements :

- 1. No manager is leader.
- 2. All leaders are executives.

Conclusions :

- 1. No manager is an executive.
- 2. No executive is a manager.

(A) Only conclusion 1 follows.

(B) Only conclusion 2 follows.

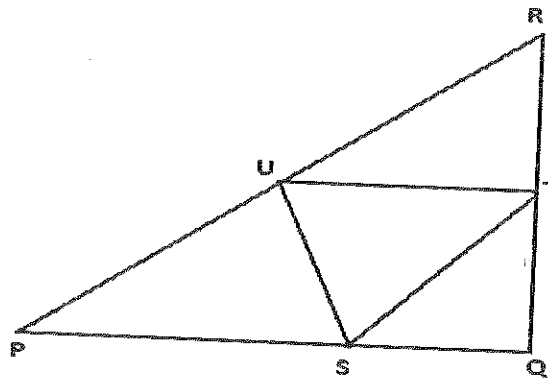
(C) Neither conclusion 1 nor 2 follows.

(D) Both conclusions 1 and 2 follows.

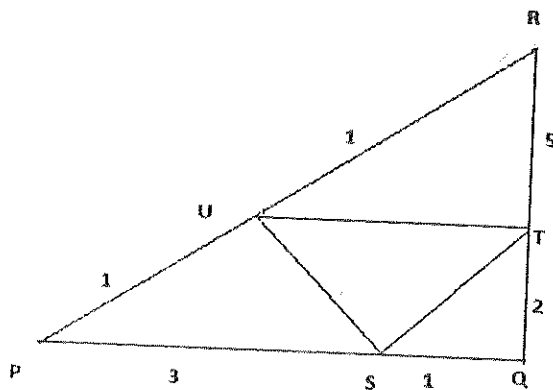
Answer (C)

Q.63 In the given figure angle Q is a right angle, $PS:QS = 3:1$, $RT:QT = 5:2$ and $PU:UR = 1:1$. If area of triangle QTS is 20 cm^2 , then the area of triangle PQR in cm^2 is _____

Answer : 280.



Solution .



In similar triangles ΔPQR and ΔQTS

$$\frac{QS}{PQ} = \frac{QT}{RQ}$$

$$\text{Area of } \triangle QTS = \frac{1}{2} \times QT \times QS$$

$$20 = \frac{1}{2} \times QT \times QS \dots\dots\dots(1)$$

$$\text{Area of } \triangle PQR = \frac{1}{2} \times PQ \times QR$$

$$\frac{QS}{PQ} = \frac{1}{1+3} = \frac{1}{4}$$

$$\therefore QS = \frac{PQ}{4}$$

$$\text{And } \frac{QT}{RQ} = \frac{2}{2+5} = \frac{2}{7}$$

$$\therefore QT = \frac{2 \times RQ}{7}$$

From equation (i),

$$20 = \frac{2}{4 \times 7} \left(\frac{1}{2} \times PQ \times QR \right)$$

$$20 = \frac{1}{14} (\text{area of } \triangle PQR)$$

Hence,

$$\text{Area of } \triangle PQR = 280 \text{ cm}^2$$

- Q.64** Right triangle PQR is to be constructed in the xy-plane so that the right angle triangle is at P and line PR is parallel to the x axis. The x and y coordinates of P,Q and R are to be integers that satisfy the inequalities : $-4 \leq x \leq 5$ and $6 \leq y \leq 16$. How many different triangles could be constructed with these properties?

- (A) 110 (B) 1,100 (C) 9,900 (D) 10,000

Answer (C)

Solution ,

$$X_1 \rightarrow -4 \leq X_1 \leq 5$$

$$Y_1 \rightarrow 6 \leq Y_1 \leq 16$$

$$X_2 \rightarrow 9 \text{ chairs } (\because X_1 \neq X_2)$$

$$Y_2 \rightarrow 10 \text{ chairs } (\because Y_1 \neq Y_2)$$

$$\therefore \text{Total triangles} = 10 \times 11 \times 9 \times 10 = 9900$$

- Q.65 A coin is tossed thrice. Let X be the event that head occurs in each of the first two tosses. Let Y be the event that a tail occurs on the third toss. Let Z be the event that two tails occur in three tosses. Based on the above information, which one of the following statement is TRUE ?
- (A) X and Y are not independent (B) Y and Z are dependent
 (C) Y and Z are independent (D) X and Z are independent

Answer (B)